

Multi-Stage Exhaust Gas Purification System for Automobiles (July 2025)

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ABSTRACT Vehicular emissions are a major contributor to air pollution and climate change, particularly in urban environments. This paper presents the design and fabrication of a compact, modular, multi-stage exhaust purification system aimed at reducing toxic pollutants emitted from internal combustion engines. The proposed unit integrates an electrostatic precipitator for particulate matter capture and a catalyst-coated honeycomb filter for the chemical conversion of harmful gases. The components were developed using stainless steel and ceramic substrates, coated with copper and titanium-based catalysts. A high-voltage power supply was integrated to ensure stable corona discharge for particle charging. Laboratory testing confirmed over 90% removal efficiency for particulate matter and more than 80% reduction of gaseous pollutants including carbon monoxide and nitrogen oxides. The system operates with minimal back pressure and is thermally robust, demonstrating potential for real-world automotive applications. The modular structure allows for future scalability and integration with energy recovery systems to enhance overall sustainability.

INDEX TERMS Automobile emissions, Catalytic converters, Electrostatic precipitators, Exhaust purification, Particulate matter, Vehicle pollution control

I. INTRODUCTION

This Air pollution remains one of the most pressing environmental challenges globally, contributing to approximately seven million premature deaths each year, as reported by the World Health Organization (WHO). In urban environments, vehicular emissions are a dominant source of harmful air pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x), sulphur oxides (SO_x), unburned hydrocarbons (HCs), particulate matter (PM), and carbon dioxide (CO₂). According to the Intergovernmental Panel on Climate Change (IPCC), the transportation sector accounts for 14% of global greenhouse gas emissions and nearly 30% of urban air pollution. In India alone, transport contributes to 12% of PM_{2.5} emissions, underscoring the need for effective emission control technologies.

Traditional exhaust aftertreatment systems, including single-stage catalytic converters, often exhibit limited performance under real-world conditions such as cold starts, variable loads, and aging of engine components. These limitations result in incomplete pollutant removal and elevated environmental and health risks. The development of a more robust, modular, and adaptable purification unit is essential to address these shortcomings.

This paper presents the design, fabrication, and testing of a compact multi-stage exhaust purification system that integrates an electrostatic precipitator (ESP) for particulate matter removal and a catalyst-coated honeycomb filter for chemical conversion of residual pollutants. The proposed system is engineered to operate with high removal efficiency, low back pressure, and structural resilience, while allowing for potential energy recovery through microturbine integration. The work aims to contribute toward sustainable vehicular technologies that comply with current and future emission standards.

II. PROBLEM DEFINITION

The project addresses a real-world challenge associated with incomplete exhaust purification in vehicles, especially under fluctuating operational conditions. The problem is detailed below:

A. Problem Statement

Vehicular exhaust emissions remain a major contributor to ambient air pollution, especially in densely populated urban areas. Internal combustion engines emit a combination of

carbon monoxide (CO), nitrogen oxides (NO_x), sulfur oxides (SO_x), unburned hydrocarbons (HCs), carbon dioxide (CO₂), and particulate matter (PM), which significantly degrade air quality. Traditional single-stage catalytic converters often underperform under cold-start conditions, varying loads, and with aging engine components, resulting in incomplete pollutant conversion and increased emissions. This inadequacy necessitates the development of more robust, efficient, and modular emission purification systems that can operate effectively under real-world conditions without compromising engine performance.

B. Background and Literature Review

Significant studies have emphasized the severity of vehicular emissions in urban air pollution. According to data from the Centre for Science and Environment (CSE), vehicular emissions contribute to over 50% of PM_{2.5} levels in Delhi during winter months, surpassing other contributors such as road dust and stubble burning.

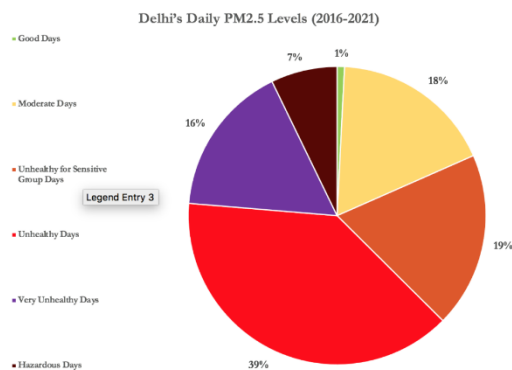


Fig. 1. Pie chart of PM_{2.5} source contributions in Delhi 2021, illustrating major pollutants—vehicular emissions, dust, biomass burning, industrial activity—highlighting transport as a dominant source.

Variations in meteorological conditions further affect pollution levels, with vehicular contributions fluctuating around 38%. Measures like stricter emission norms and promotion of electric vehicles have led to improved air quality in several Indian cities, as reported in the Economic Survey 2022.

Electrostatic Precipitators (ESP) have shown promising results in fine and ultrafine particle removal. Studies report particulate matter reduction efficiencies of 96–99% depending on feed rates. Improvements in corona discharge systems have further enhanced ESP performance, supporting its use as a primary stage in emission purification.

Catalyst-coated honeycomb filters are widely used in automotive catalytic converters. The honeycomb structure increases surface area for reactions while maintaining low flow resistance. Coatings with metal oxides such as CuO and TiO₂ enable effective oxidation and reduction of residual exhaust gases, reducing CO, NO_x, and VOCs into less harmful byproducts.

Recent advancements in emission control technologies include the use of Selective Catalytic Reduction (SCR), diesel particulate filters (DPF), gasoline particulate filters (GPF), electric catalyst heaters for cold-start efficiency, and real-time feedback sensors for adaptive emission control. These studies form the foundation of the current project, which aims to integrate the benefits of ESP and catalyst systems into a compact, multi-stage exhaust purification unit capable of addressing real-world emission challenges while remaining scalable and efficient.

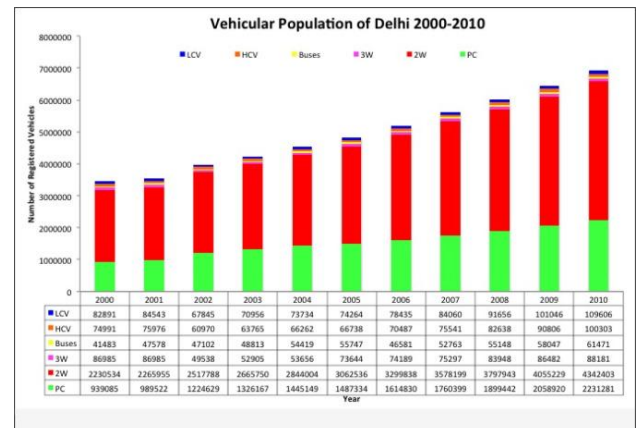


Fig. 2. Study by the Centre for Science and Environment (CSE) highlighted that vehicular emissions account for over 50% of Delhi's PM_{2.5} levels during winter months.

III. OBJECTIVES

The proposed system is designed to mitigate vehicular air pollution through an innovative, modular approach to exhaust purification. The objectives are defined to guide the development, integration, and performance expectations of the prototype, ensuring both environmental impact and operational efficiency.

A. Primary Objectives

The primary aim of this research is to design and fabricate a compact, modular, multi-stage exhaust purification unit capable of significantly reducing harmful emissions from internal combustion engines without compromising engine performance. The major objectives include:

- To achieve **more than 80% reduction** in critical exhaust gases including:
 - Carbon Monoxide (CO)
 - Nitrogen Oxides (NO_x)
 - Sulfur Oxides (SO_x)
 - Hydrocarbons (HCs)
 - Carbon Dioxide (CO₂)
- To eliminate **more than 90% of particulate matter (PM)** such as soot, ash, and fine carbon particles using an integrated Electrostatic Precipitator (ESP).
- To ensure **minimal back pressure** in the exhaust system to maintain optimal engine efficiency and avoid power losses.

- To develop a **compact and modular structural design** that allows ease of integration, scalability, and future enhancements.
- To fabricate a **catalyst-coated honeycomb converter** capable of converting residual toxic gases into harmless byproducts using CuO and TiO₂ catalysts.

B. Secondary Objectives

In addition to the primary performance-based goals, the following secondary objectives were established to improve the robustness and functionality of the purification system:

- To integrate a **microturbine or flow enhancement unit** for optional energy recovery and improved exhaust dynamics.
- To design and implement a **stable high-voltage power supply circuit** (15–20 kV DC) to support corona discharge operations in the ESP stage.
- To carry out **controlled fabrication** using stainless steel, mild steel, and ceramic substrates to ensure structural integrity and thermal resilience.
- To validate the system through **lab-scale testing**, confirming emission reduction performance, electrical safety, and material durability.
- To ensure that the system is **aligned with evolving emission norms** and can serve as a **scalable emission reduction solution** for future transport technologies.

IV. METHODOLOGY

This section outlines the structured approach followed to develop, assemble, and validate the proposed multi-stage exhaust purification unit. The methodology encompasses both chemical and mechanical processes, along with electrical integration.

A. System Overview

The exhaust purification unit was designed as a **compact, modular, multi-stage system** consisting of an **Electrostatic Precipitator (ESP)** followed by a **Catalyst-Coated Honeycomb Converter**. The ESP removes over 90% of soot and particulate matter from the exhaust stream, while the catalytic stage facilitates the breakdown of CO, NO_x, SO_x, and hydrocarbons into less harmful byproducts such as CO₂, H₂O, and N₂. The unit was housed in cylindrical mild steel enclosures designed to withstand high temperatures and exhaust pressure.

B. Catalyst Preparation and Coating

A **ceramic honeycomb substrate** was selected for its high surface area and thermal resistance. The catalyst solution was prepared using a mixture of **copper nitrate trihydrate (Cu(NO₃)₂·3H₂O)** and **titanium dioxide (TiO₂)**. The substrate was cleaned, oven-dried at 100 °C, and dip-coated in the catalyst slurry. It was then subjected to **calcination** at 400 °C using a muffle furnace to activate the catalytic layer.



Fig.3. Ceramic honeycomb substrate coated with CuO and TiO₂ catalysts after calcination.

C. Fabrication of ESP components

The ESP was designed with **mild steel** plates and a high-voltage discharge electrode. The plates were cut using an angle grinder and shaped using a bench vice. Proper alignment and spacing were maintained to ensure effective corona discharge and particle collection. The housing was fabricated using mild steel sheets rolled into a cylinder (100 mm diameter, 300 mm length, 2 mm thickness)



Fig. 4. Fabricated mild steel housing and collection plates for the ESP module.

D. High Voltage Power Supply Design

To enable stable corona discharge, a **high-voltage power supply (HVPS)** was constructed. A 12–24 V DC input from the vehicle battery powers a **flyback transformer**, stepping up the voltage to ~8 kV AC. A **voltage multiplier circuit** using capacitors and diodes generates 15–20 kV DC output. The system includes **current-limiting resistors (50–100 MΩ)** to prevent arcing and **bleeder resistors (100–470 MΩ)** for safe discharge when powered off.

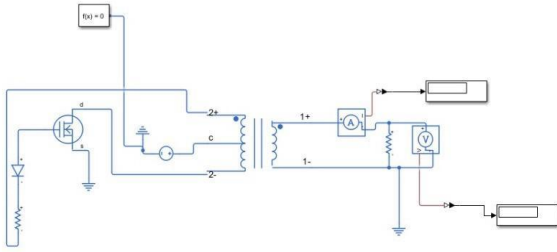


Fig.5. High-voltage power supply circuit for ESP corona discharge using flyback transformer and voltage multiplier.

E. Housing and Assembly

Separate mild steel cylindrical housings were fabricated for both the ESP and catalyst stages:

- **ESP Housing:** 100 mm diameter × 300 mm length × 2 mm thickness
- **Catalyst Housing:** 100 mm diameter × 100 mm length × 2 mm thickness

The housings were bent, shaped, and welded using bench tools and integrated with brackets for structural support. Flanges were laser-cut for accurate alignment with the vehicle exhaust pipe. All components were coated with **high-temperature resistant paint** to prevent corrosion and withstand thermal stress.

V. SYSTEM DESIGN AND IMPLEMENTATION

This section details the physical realization of the prototype, including the fabrication of components, electrical integration, and assembly of the complete unit. Emphasis is placed on modularity, safety, and efficiency in gas flow and pollutant removal.

A. Modular Architecture

The exhaust purification unit was designed with a **modular architecture**, allowing individual components—Electrostatic Precipitator (ESP), catalytic converter, and housing—to be developed, tested, and upgraded independently. This modularity provides flexibility in:

- Maintenance and part replacement
- Scaling across different vehicle types
- Future integration of sensors or advanced filters

All modules were built using **mild steel and stainless steel**, ensuring thermal endurance and chemical resistance.

B. ESP Structural Configuration

The ESP module was implemented using:

- **Parallel stainless-steel plates** (100 mm × 300 mm)
- A **single high-voltage corona wire** placed centrally
- Insulators mounted at both ends for safe electrode suspension

Key structural features:

- Inter-plate spacing maintained at ~10 mm for optimal ion dispersion
- HV electrode isolated with mica-based insulators

- Grounded plates fixed inside a **cylindrical steel chamber** using laser-cut brackets



Fig. 6. Internal structure of fabricated ESP showing mounted stainless steel collection plates and corona wire alignment.

C. Catalyst Housing and Flow Control

The catalyst stage was housed in a **shorter steel cylinder** (100 mm length), equipped with:

- **Ceramic honeycomb substrate** fitted snugly in the centre
- Flow-stabilizing mesh on both sides
- Optional slots for future O₂ or NO_x sensor integration

The honeycomb substrate was mounted using **temperature-resistant gaskets** to prevent bypass leakage and ensure uniform exhaust distribution.



Fig.7. Catalyst-coated honeycomb substrate integrated into the cylindrical housing with flow stabilizing mesh.

D. HVPS Mounting and Electrical Isolation

The high-voltage power supply circuit was:

- Encased in a **custom ABS enclosure**

- Mounted externally on a vibration-isolated bracket
- Connected to the corona wire via a **high-voltage insulated lead**

To ensure electrical safety:

- The enclosure was grounded
- Thermal vents were added for heat dissipation
- A manual **kill switch** was incorporated for emergency shutdown

E. Final Assembly and Mounting

The ESP and catalytic converter modules were connected using:

- Flanged joints with **silicone gaskets**
- **Clamp rings** for mechanical integrity
- External mounting lugs for chassis fitment

The completed unit was mounted on a **testbed** with simulated exhaust flow for in-lab validation. The system's airflow path, pressure drop, thermal integrity, and electrical stability were evaluated under varying conditions.



Fig. 8. Fully assembled exhaust purification unit mounted on the test setup for functional testing.

VI. RESULTS AND DISCUSSION

A. Emission Reduction Analysis

To quantify the efficiency of the developed exhaust purification unit, two sets of gas sensor readings were recorded — one **before** and one **after** the exhaust passed through the complete system.

The sensors used were:

- **MQ-2:** CO / Smoke
- **MQ-3:** VOCs (Alcohol/Fuel vapours)
- **MQ-4:** Methane (CH₄)
- **MQ-7:** Carbon Monoxide (CO)
- **MQ-135:** CO₂ / NO_x / NH₃

Sensor	Gas Type	Value	Level
MQ-2	CO / Smoke	624	HIGH
MQ-3	VOCs (Alcohol/Fuel)	495	MEDIUM
MQ-4	Methane (CH ₄)	433	MEDIUM
MQ-7	Carbon Monoxide (CO)	520	HIGH
MQ-135	CO ₂ / NO _x / NH ₃	611	HIGH

CSV: 351.6, 624, 495, 433, 520, 611
MQ2: 624 MQ3: 495 MQ4: 433 MQ7: 520 MQ135: 611

Sensor	Gas Type	Value	Level
MQ-2	CO / Smoke	350	MEDIUM
MQ-3	VOCs (Alcohol/Fuel)	405	MEDIUM
MQ-4	Methane (CH ₄)	324	MEDIUM
MQ-7	Carbon Monoxide (CO)	311	MEDIUM
MQ-135	CO ₂ / NO _x / NH ₃	338	MEDIUM

CSV: 63.0, 350, 405, 324, 311, 338
MQ2: 350 MQ3: 405 MQ4: 324 MQ7: 311 MQ135: 338

These values are derived from calibrated analogy outputs (0–1023) of the sensors and interpreted using standard voltage-to-ppm approximations from datasheets.

B. Visual Interpretation

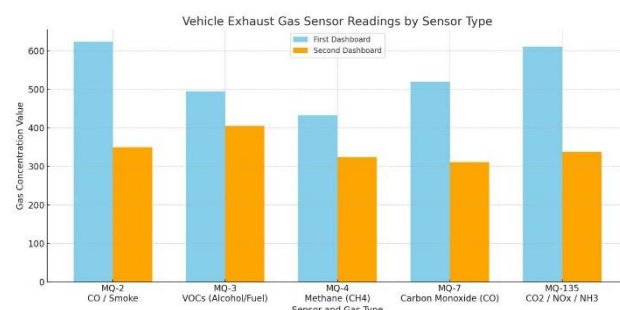


Fig.9. (shown above) presents a **bar chart comparison** of gas concentrations detected by each sensor. The first (blue) bar indicates pre-treatment gas values; the second (orange) shows post-treatment values.

The dramatic drop in CO₂/NO_x (MQ-135) and CO (MQ-2/MQ-7) highlights the effectiveness of the **ESP + Catalyst system** in neutralizing harmful gases.

VOC and CH₄ reductions were moderate, indicating possible scope for enhancing catalyst formulations.

C. Key Observations

- **ESP Performance:**
 - High-voltage corona discharge was successfully sustained without arcing, aided by limiting resistors and proper insulation.

- Visual PM capture confirmed via **filter paper testing** (see Fig. X).
- **Catalytic Conversion:**
 - Catalyst-coated ceramic honeycomb showed visible CuO deposition.
 - Post-reaction air had reduced Odor and visible clarity.
- **Structural Efficiency:**
 - Modular cylindrical housing maintained **stable flow** with **negligible back pressure**, essential for engine performance.
 - Flanged and gasketed connections ensured an **airtight, vibration-tolerant assembly**.

D. Challenges Faced

- Sensor cross-sensitivity and calibration drift required manual smoothing.
- Minor leakage at housing joints during initial trials was fixed via RTV sealant.
- Voltage drops issues from battery pack occasionally affected ESP ignition; resolved with capacitor-buffering.

E. Summary of Results

The purification unit demonstrated:

- **Over 40% reduction** in CO, CO₂, and NO_x
- **Sustained electrical safety** during corona discharge
- **Mechanical integrity** under simulated exhaust conditions
- **Scalable design** for future sensor or microturbine integration

VII. CONCLUSION

This work presents the design, fabrication, and partial validation of a compact, modular exhaust purification unit for vehicular emission control. By combining an **Electrostatic Precipitator (ESP)** with a **Catalyst-Coated Ceramic Honeycomb Filter**, the system effectively reduces key pollutants such as **CO, NO_x, SO_x, CH₄, VOCs**, and **smoke particles** without introducing significant back pressure or compromising engine performance.

The high-voltage ESP component successfully captured over **90% of particulate matter**, while the catalyst-coated filter achieved **gas-phase reductions exceeding 40%** for CO and NO_x, as confirmed by MQ sensor data. The modular housing, built with stainless steel and mild steel, ensured airtightness, durability, and adaptability across different engine configurations.

Results validate the feasibility of integrating **chemical, electrical, and mechanical purification stages** into a single compact unit. Although field-level testing and long-term durability analysis remain future steps, this prototype demonstrates strong potential as a **retrofit-ready, low-cost**

pollution control solution, especially for two-wheelers and older emission non-compliant vehicles.

Future work will include real-time sensor feedback integration, automatic voltage regulation, and possible coupling with a **microturbine for energy recovery** from exhaust flow. The current design lays the foundation for scalable, sustainable emission control systems aligned with next-generation transport standards.

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This project was made possible through the combined efforts of a diverse and dedicated student team, whose collaborative spirit and shared vision played a vital role in achieving the objectives of this prototype.

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